

AEROFATIGUE OPERATIONAL SAFETY SYSTEM

COMPREHENSIVE USER MANUAL & SYSTEM OPERATIONS GUIDE

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Firestore collections (aerofatigue_personnel, aerofatigue_logs, aerofatigue_config) across all devices.

1. SYSTEM OVERVIEW & CORE OBJECTIVES

AeroFatigue is a specialized Aviation Operational Fatigue Risk Management System (FRMS) designed to

continue without interruption and automatically synchronize when connectivity is restored.

2. DUAL OPERATIONAL ARCHITECTURE & USER ROLES

- [A] AIR CREW DIVISION (Pilots, Navigators, Tactical System Operators, Flight Engineers)**
 - Focuses on sortie duration, flight intensity, instrument flight (IFR), Night WOCL operations, and rest cycles.
- [B] GROUND CREW DIVISION (Line Maintenance, Defect Rectification, Avionics, AMCO, Unit Duty)**
 - Monitors hands-on defect rectification, hourly/calendar servicing, flight servicing (AFS/BFS/TRS), and duty chief.
- [C] COMMANDER / ADMIN MASTER CONSOLE**
 - Unified fleet-wide command dashboard aggregating all Air Crew and Ground Crew fatigue indices.
- [D] INDIVIDUAL CREW MEMBER DASHBOARD**
 - Protected personal portal with 4-digit PIN for logging duty entries, viewing graphs, and reviewing history.

3. CLOUD PERSISTENCE & OFFLINE ARCHITECTURE

- 1. Real-Time Google Firebase Firestore Cloud Sync:**

Every personnel record, fatigue check-in, profile update, and audit log is automatically synchronized with cloud
- 2. Resilient Offline Local Caching:**

The application maintains a high-speed local browser cache. If internet or server connection drops, all operations
- 3. Firebase Multi-Factor Authentication:**

Supports secure Email/Password authentication and one-click Google Sign-In for authorized military and ops staff.

Section 2: Empirical Risk Formulation, Weighting Factors & Algorithmic Computations

THE 5-FACTOR FATIGUE ACCUMULATION ALGORITHM (0 TO 100 POINTS)

1. Sleep Debt Penalty Factor (0 to 30 Points):

Calculated from prior 24-hour sleep duration with progressive non-linear escalation below 6 hours.

2. Duty Period Duration Factor (0 to 25 Points):

Derived dynamically from Duty Start Time to Duty End Time:

3. Circadian Rhythm & Shift Window (0 or 20 Points):

- Day Shift (06:00 to 22:00): 0 Points (Natural biological diurnal activity)

- Scheduled Servicing: Mandatory hourly-based and calendar-based major scheduled servicing.

4. Consecutive Duty Days Progression (0 to 10 Points):

Auto-calculated from historical shift dates to capture cumulative chronic fatigue:

5. Workload Intensity Factor (0 to 10 Points):

- Low (0 pts) | Medium (+5 pts) | High (+10 pts) -- Governed by Aircraft Workload Manual

WORK INTENSITY SQUADRON CLASSIFICATION REFERENCE

[+] LOW-INTENSITY WORK (0 Points)

- Documentation & Records: Form filling, system logs, completing job cards, maintenance records.

[+] MEDIUM-INTENSITY WORK (+5 Points)

- Flight Servicing: AFS, BFS, TRS, and BTWS operational aircraft preparation activities.

[+] HIGH-INTENSITY WORK (+10 Points)

- Defect Rectification: Direct technical troubleshooting, diagnosis, defect isolation, component replacement.

MASTER FORMULA: Score = Sleep (0-30) + Duty (0-25) + Shift (0/20) + Consec (0-10) + Workload (0-10)

Final Cartesian score is clamped strictly between [0 and 100] and categorized into 4 Operational Risk Bands.

CARTESIAN RISK BANDS & SUPERVISORY PROTOCOLS

Section 3: Four-Tier Operational Risk Matrix, Thresholds & Mandatory Actions

FOUR-TIER OPERATIONAL RISK BANDS & MANDATORY MITIGATION	
	?? 1. LOW RISK (Score: 0 to 24 Points) - GREEN BAND - Status: Personnel fully fit for flight, maintenance, and tactical operations. Optimal mental alertness.
	24-hour restorative rest period. ?? 2. MODERATE RISK (Score: 25 to 49 Points) - YELLOW BAND - Status: Noticeable fatigue onset. Minor degradation in reaction times and sustained vigilance.
	?? 3. HIGH RISK (Score: 50 to 74 Points) - ORANGE BAND - Status: High cumulative fatigue. Divided attention deficits and elevated risk of procedural slips.
	independent inspection by a designated Flight Safety Officer or Line Chief. ?? 4. CRITICAL RISK (Score: 75 to 100 Points) - RED BAND - Status: Severe exhaustion. Unacceptable probability of catastrophic operational error.

MANDATORY OPERATIONAL SAFETY DIRECTIVES (Aviation FRMS Standards)	
	1. Maximum Continuous Duty Shift Limit: Standard duty periods must not exceed 12 continuous hours. Shifts extending beyond 12 hours require
	2. Night Duty (WOCL) Consecutive Shifts Limit: Personnel must not be assigned to more than three (3) consecutive night duty shifts without an intervening
	3. In-Flight Relief Crew Rotation: Mandatory in-flight relief crew rotation is enforced when mission sortie duration exceeds 7.5 flight hours.
	4. Duplicate Inspection Protocol for Maintenance: Any defect rectification performed by ground crew in High Risk (Orange) state requires mandatory duplicate

USER INTERFACE & STEP-BY-STEP OPERATOR GUIDE

? Security: Change Admin Password, inspect Firebase connection status, and audit active crew.

Section 4: Navigation, Personnel Management, Duty Recording & Admin Controls

1. NAVIGATING THE OPERATIONAL PORTAL & SQUADRON ROSTER

A. Section Switching & Search:

- Click [? AIR CREW] or [? GROUND CREW] tab to switch rosters.

B. Adding & Editing Personnel:

- Click [+ ADD PERSON] to register a new crew member. Provide Name, Designation, Unit, and 4-digit PIN.

? Master History: Searchable cross-squadron duty logs with sorting and filtering.

C. Accessing Individual Crew Dashboard:

- Click on any crew member card to view their shift history and logs (Default on 1234).

2. STEP-BY-STEP DUTY RECORDING & DASHBOARD TOOLS

Step 1 - Date & Synchronized Live System Time:

- Date and record time are automatically synchronized with system current time. Editable if logging past shifts.

Step 2 - Duty Cycle, Shift Category & Intensity:

- Set Duty Start and End Times (Duration is calculated dynamically in real time).

? Fleet Overview: Real-time risk distribution metrics, section breakdowns, and high-risk alerts.

Step 3 - Rest & Operational Notes:

- Input Sleep Duration in hours obtained prior to duty (0.5h step slider).

Interactive Cartesian Graph & History Log:

- Graph tab renders SVG Cartesian fatigue curves across time with hover tooltips and colored risk bands.

3. COMMANDER ADMIN CONSOLE & REFERENCE CENTER

? Admin Console Login: Click [??? ADMIN COMMAND] (Default ID: admin | Pass: admin123).